MCE5919: Gaussian Process for Machine Learning and Signal Processing

Lecture 2

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Overview

1. Preliminaries

2. Linear Regression

3. Logistic Regression

Notation

- Vectors in bold lower-case, e.g., $\boldsymbol{x}$; matrices in bold upper-case, e.g., $\mathbf{X}$
- $\mathbf{I}$: identity matrix; $\mathbf{0}$: all-zero vector/matrix
- $\mathrm{tr}(\cdot)$: trace; $(\cdot)^T$: transpose; $(\cdot)^{-1}$: inverse
- $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$: Gaussian with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$

Gaussian Distribution

Univariate: If $Y \sim \mathcal{N}(\mu, \sigma^2)$,

$$p(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\,\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right).$$

Multivariate: If $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$p(\mathbf{y}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{n/2}\,|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{y}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{y}-\boldsymbol{\mu})\right).$$

- $E[\mathbf{y}] = \boldsymbol{\mu}$, $\mathrm{Cov}(\mathbf{y}) = \boldsymbol{\Sigma}$

Chi-square Distribution

If $Z_1, \ldots, Z_r \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, then

$$\chi^2_r \equiv \sum_{i=1}^{r} Z_i^2 \sim \text{Chi-square}(r), \quad E[\chi^2_r] = r,\ \mathrm{Var}[\chi^2_r] = 2r.$$

In regression: With Gaussian errors, a properly scaled residual sum of squares follows a chi-square distribution and underpins inference for σ^2.

Bernoulli Distribution

Let $Y \in \{0, 1\}$ be a Bernoulli random variable with parameter $p \in (0, 1)$.

$$P(Y = y; p) = p^y (1 - p)^{1-y}, \quad y \in \{0, 1\}.$$

- $E[Y] = p$, $\mathrm{Var}(Y) = p(1 - p)$
- For i.i.d. samples $\{y_i\}_{i=1}^n$, the likelihood is $L(p) = \prod_{i=1}^n p^{y_i}(1 - p)^{1-y_i}$

Unbiased Estimators & Covariance

Unbiasedness: An estimator $\hat{\theta}$ is unbiased if $E[\hat{\theta}] = \theta$.

Sample Covariance: For data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ with rows $\mathbf{x}_i$ and mean $\bar{\mathbf{x}}$, the unbiased covariance estimate is

$$\hat{\mathbf{\Sigma}} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T.$$

Estimator Covariance: For LS in linear regression, $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$.

Fisher Information & CRLB

Fisher Information (scalar):

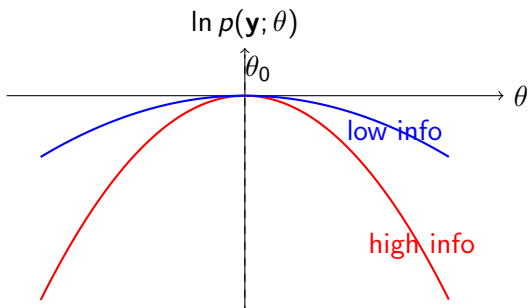
$$I(\theta) = E\left[\left(\frac{\partial}{\partial\theta}\ln p(\mathbf{y};\theta)\right)^2\right] = -E\left[\frac{\partial^2}{\partial\theta^2}\ln p(\mathbf{y};\theta)\right].$$

Vector case: $\mathbf{I}(\boldsymbol{\theta}) = E\left[\nabla_{\boldsymbol{\theta}} \ln p \, \nabla_{\boldsymbol{\theta}} \ln p^T\right]$.

CRLB: For any unbiased estimator, $\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{I}(\boldsymbol{\theta})^{-1}$; asymptotically, ML attains this bound under regularity conditions.

Fisher Information: Intuition

- Measures how sensitive the likelihood is to parameter changes: steeper log-likelihood near true $\theta \Rightarrow$ more information.
- For scalar θ, larger curvature $\left|\frac{\partial^2}{\partial \theta^2} \ln p\right| \Rightarrow$ larger $I(\theta)$.



Fisher Information: Examples

Bernoulli(p): $Y \in \{0, 1\}$, $P(Y = 1) = p$.

$$I(p) = \frac{1}{p(1-p)}, \quad I_n(p) = \frac{n}{p(1-p)} \text{ for i.i.d. samples.}$$

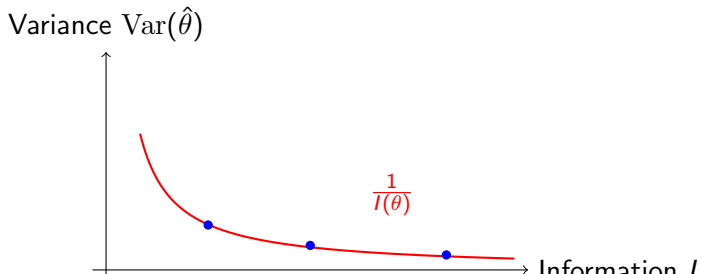
Gaussian Mean: $Y \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2.

$$I(\mu) = \frac{n}{\sigma^2}, \quad \text{larger } n \text{ or smaller } \sigma^2 \Rightarrow \text{more information.}$$

Reading: Information increases when data are more precise (small noise) or abundant (large n).

CRLB: Intuition

- For any unbiased estimator $\hat{\boldsymbol{\theta}}$, its covariance obeys $\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{I}(\boldsymbol{\theta})^{-1}$.
- 1D case: $\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$ — the variance floor set by information content.
- More information $\Rightarrow$ tighter bound (unbiasedness and regularity conditions assumed).



CRLB in Practice

- **Bernoulli**(p): $\hat{p} = \frac{1}{n}\sum y_i$ is unbiased with $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n} = \frac{1}{I_n(p)}$ — attains CRLB.
- **Gaussian Mean**: sample mean $\bar{Y}$ is unbiased with $\mathrm{Var}(\bar{Y}) = \frac{\sigma^2}{n} = \frac{1}{I(\mu)}$ — attains CRLB.
- **Linear Regression (Gaussian errors)**: MLE/LS $\hat{\boldsymbol{\beta}}$ has $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$, which equals CRLB under regularity.

Optimization: Convex Quadratic

Consider the unconstrained convex quadratic

$$\min_{\boldsymbol{x}} \ \frac{1}{2}\boldsymbol{x}^T\mathbf{Q}\boldsymbol{x} + \boldsymbol{c}^T\boldsymbol{x} + r, \quad \mathbf{Q} \succ 0.$$

Solution: Set gradient to zero,

$$\mathbf{Q}\,\boldsymbol{x} + \boldsymbol{c} = \mathbf{0} \ \Rightarrow \ \boldsymbol{x}^\star = -\mathbf{Q}^{-1}\boldsymbol{c}.$$

Link to LS: With $\mathbf{Q} = \mathbf{X}^T\mathbf{X}$ and $\boldsymbol{c} = -\mathbf{X}^T\mathbf{y}$, we recover $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$.

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Multiple Linear Regression Model

A multiple linear regression model is given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon$$

where

- unknown model parameters $\beta_j, j = 1, 2, ..., k$ are often called regression coefficients
- $x_j, j = 1, 2, ..., k$ are the inputs/regressors and y is the output/response
- inputs $x_j, j = 1, 2, ..., k$ are deterministic and precisely known
- ε is random error term with zero mean and unknown variance σ^2

Remark: No specific error distribution is assumed herein

Multiple Linear Regression Model

Assuming *n* observations serving as training data, the above regression model can be written in a matrix form as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $(i \in \{1, 2, ..., n\})$

$$\begin{aligned}
\mathbf{y} &= [y_1, y_2, ..., y_n]^T, \\
\mathbf{x}_i &= [1, x_{i,1}, x_{i,2}, ..., x_{i,k}]^T, \\
\mathbf{X} &= [\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_n]^T, \\
\boldsymbol{\beta} &= [\beta_0, \beta_1, ..., \beta_k]^T, \\
\boldsymbol{\varepsilon} &= [\varepsilon_1, \varepsilon_2, ..., \varepsilon_n]^T.
\end{aligned}$$

LS Parameter Estimation: $\boldsymbol{\beta}$

Derive the least-squares (LS) estimator of $\boldsymbol{\beta}$:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} S(\boldsymbol{\beta}) \triangleq (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

The LS estimator (in matrix form) is given by

$$\frac{\partial S}{\partial \boldsymbol{\beta}}|_{\hat{\boldsymbol{\beta}}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}$$

If $(\mathbf{X}^T\mathbf{X})^{-1}$ exists, we finally have

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

LS Parameter Estimation: σ^2

As a result, the vector of fitted values $\hat{\mathbf{y}} = [\hat{y}_1, \hat{y}_2, ..., \hat{y}_n]^T$ is computed by

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$$

where the matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is called the hat matrix.

Consequently, the vector of residuals, $\mathbf{e} = [e_1, e_2, ..., e_n]^T$, is computed as

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}$$

It can be easily verified that both $\mathbf{H}$ and $\mathbf{I} - \mathbf{H}$ are idempotent matrices.

An Estimator of σ^2

Residual/error sum of squares is defined as

$$SS_{Res} = \sum_{i=1}^{n} e_i^2 = \mathbf{e}^T \mathbf{e}$$

An estimator of σ^2 is given by

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n - p}$$

Properties of the LS Estimator: $\hat{\sigma}^2$

It is assumed that the data is truly from the model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$.

It can be proven that :

1. An unbiased estimator of σ^2 is given by

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n-p}$$

2. Under the assumption that $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, SS_{Res}/σ^2 follows χ^2_{n-p} distribution, where $p = k + 1$.[1]

[1] If $Z_1, \ldots, Z_n$ are independent identically distributed (i.i.d.), standard normal random variables, then $\sum_{t=1}^{n} \left(Z_t - \bar{Z}\right)^2 \sim \chi^2_{n-1}$ where $\bar{Z} = \frac{1}{n}\sum_{t=1}^{n} Z_t$.

1. To prove $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p}$ is an unbiased estimator of σ^2, we need:

Theorem 1

Let $\mathbf{A}$ be a $k \times k$ matrix of constants and $\mathbf{y}$ be a $k \times 1$ multivariate random vector with mean $\boldsymbol{\mu}$ and non-singular covariance matrix $\boldsymbol{\Sigma}$. Let U be the quadratic form defined by $U = \mathbf{y}^T\mathbf{A}\mathbf{y}$, then $E(U) = tr(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$.

2. To prove under another assumption that the error $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, SS_{Res}/σ^2 follows χ^2_{n-p} distribution, we need:

Theorem 2

Let $\mathbf{A}$ be a $k \times k$ idempotent matrix of constants with rank r and $\mathbf{y}$ be a $k \times 1$ multivariate Gaussian random vector with zero-mean $\boldsymbol{\mu}$ and non-singular covariance matrix $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$. Let U be the quadratic form defined by $U = \mathbf{y}^T\mathbf{A}\mathbf{y}$, then $\frac{U}{\sigma^2} \sim \chi^2_r$.

Properties of the LS Estimator: $\hat{\boldsymbol{\beta}}$

Derive/Prove that:

1. $E\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$
2. $\mathrm{Cov}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2\mathbf{C} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$, where the variance of $\hat{\beta}_j$ is $\sigma^2 C_{jj}$ and the covariance between $\hat{\beta}_i$ and $\hat{\beta}_j$ is $\sigma^2 C_{ij}$
3. LS estimator $\hat{\boldsymbol{\beta}}$ is the best linear unbiased estimator (BLUE) of $\boldsymbol{\beta}$ (Gauss-Markov Theorem)

Maximum-Likelihood (ML) Estimation

LS estimation

The LS estimation does not assume any specific distribution of the random error ε

ML estimation

The ML estimation does assume full statistics of the random error ε

Remark: The idea of the (Frequentist) ML estimation is to find the point estimate of $\boldsymbol{\beta}$ that maximizes the likelihood function, $p(\mathbf{y}; \boldsymbol{\beta})$, i.e., the value that makes the observed data most probable!

ML Estimator vs. LS Estimator

For the multiple linear regression model with i.i.d. Gaussian errors, i.e., $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, the likelihood function is

$$p(\mathbf{y}; \boldsymbol{\beta}, \sigma^2) \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n) = \frac{1}{\left(\sqrt{2\pi}\sigma\right)^n} \exp\left[\frac{-(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}{2\sigma^2}\right]$$

Often, we take the logarithm of the likelihood function, short as log-likelihood, namely

$$\ln p(\mathbf{y}; \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

ML Estimator

Taking the derivative of $\ln p(\mathbf{y}; \boldsymbol{\beta}, \sigma^2)$ with respect to $\boldsymbol{\beta}$ and σ^2 respectively and setting them equal to zero, yields:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y},$$
$$\hat{\sigma}^2 = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n}$$

What is the difference between ML estimator and LS estimator?

- $\hat{\boldsymbol{\beta}}$ is obtained as a point estimate without uncertainty quantification.
- Point prediction can be made straightforwardly via (for a novel input $\mathbf{x}_*$)

$$\hat{y}(\mathbf{x}_*) = f(\mathbf{x}_*; \hat{\boldsymbol{\beta}})$$

ML: Asymptotic Efficiency

$\rightsquigarrow$ ML estimator for general model $f(\mathbf{x};\boldsymbol{\theta})$ is asymptotically optimal!

Asymptotic Properties of the ML Estimator

If $p(\mathbf{y};\boldsymbol{\theta})$ of the output $\mathbf{y}$ satisfies some regularity conditions, then the ML estimator of the unknown parameters $\boldsymbol{\theta}$ is asymptotically distributed (i.e., for very large data records) according to

$$\hat{\boldsymbol{\theta}} \overset{a}{\sim} \mathcal{N}\left(\boldsymbol{\theta}, I^{-1}(\boldsymbol{\theta})\right)$$

where $I(\boldsymbol{\theta})$ is Fisher information evaluated at the true value of the unknown parameter and $I^{-1}(\boldsymbol{\theta})$ is Cramer-Rao lower bound (CRLB) for benchmarking unbiased parameter estimators.

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Logistic Regression Model

Logistic regression is a class of regression models with

- output, y, is qualitative/categorical/discrete-valued;
- input, $\mathbf{x}$, is not constrained.

Binary Logistic Regression

- Binary logistic regression models assume that the output has two possible outcomes, generically called "is-the-case" (1) and "not-the-case" (0).
- In machine learning, this is also known as binary classification.

Multinomial Logistic Regression

Multinomial logistic regression models assume that the output has more than two possible outcomes.

Logistic Regression Model

We assume that the output y given a specific $\mathbf{x}$ is a Bernoulli random variable with the probability mass function as follows:

y	probability mass function
1	$P(y = 1; \mathbf{x}) = p$
0	$P(y = 0; \mathbf{x}) = 1 - p$

- Widely used Sigmoid model.

Logistic Regression Model

- Sigmoid function has the form

$$\sigma(z) = \frac{1}{1 + \exp(-z)} : \mathbb{R} \to \mathbb{R},\ z \in (-\infty, \infty)$$

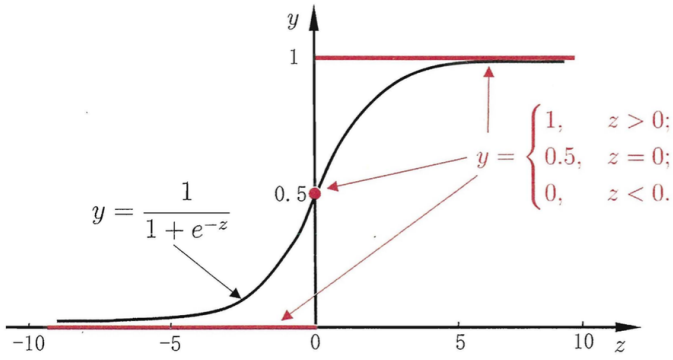


Figure: Sigmoid function (in black color) versus the unit step function (in red color).

Logistic Regression Model

Binary logistic regression model in general form:

$$p(y = 1 \,|\, \mathbf{x}) \;=\; \sigma(\mathbf{x}^T \boldsymbol{\beta}) \;=\; \frac{1}{1 + \exp(-\mathbf{x}^T \boldsymbol{\beta})}$$

$$p(y = 0 \,|\, \mathbf{x}) \;=\; 1 - \sigma(\mathbf{x}^T \boldsymbol{\beta}) \;=\; \frac{\exp(-\mathbf{x}^T \boldsymbol{\beta})}{1 + \exp(-\mathbf{x}^T \boldsymbol{\beta})}$$

- The output $y \in \{0, 1\}$ indicates the class label.
- Inputs $\mathbf{x} = [1, x_1, x_2, ..., x_k]^T$ defined similarly as before.
- Model parameters $\boldsymbol{\beta} = [\beta_0, \beta_1, ..., \beta_k]^T$ to be estimated via ML.

Question: Why can't we simply use multiple linear model for $f(\mathbf{x}; \boldsymbol{\beta})$.

Training Phase vs. Test Phase

1. Training phase: given a set of n observations in the training dataset $\mathcal{D} = \{(y_1, \mathbf{x}_1), (y_2, \mathbf{x}_2), ..., (y_n, \mathbf{x}_n)\}$, we fit the model parameters $\boldsymbol{\beta}$ and obtain the maximum-likelihood estimate $\hat{\boldsymbol{\beta}}_{\mathrm{ML}}$.
2. Test phase: given a novel input $\mathbf{x}_*$, we desire an accurate prediction on the probability of having "0" or "1" based on the trained logistic regression model.

Model Fit Using ML Estimation

Given the training dataset, the likelihood function is given as follows:

$$L(\mathbf{y}; \mathbf{X}, \boldsymbol{\beta}) = L(y_1, y_2, ..., y_n; \mathbf{X}, \boldsymbol{\beta}) = \prod_{i=1}^{n} P(y_i; \mathbf{x}_i, \boldsymbol{\beta}) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1-y_i}.$$

It is more convenient to work with the log-likelihood:

$$\begin{aligned} \ln L(\mathbf{y}; \mathbf{X}, \boldsymbol{\beta}) &= \ln \prod_{i=1}^{n} P(y_i; \mathbf{x}_i, \boldsymbol{\beta}) \\ &= \sum_{i=1}^{n} \ln(1 - p_i) + \sum_{i=1}^{n} y_i \ln \left(\frac{p_i}{1 - p_i} \right) \\ &= \sum_{i=1}^{n} y_i \mathbf{x}_i^T \boldsymbol{\beta} - \sum_{i=1}^{n} \ln \left(1 + \exp(\mathbf{x}_i^T \boldsymbol{\beta}) \right). \end{aligned}$$

Newton's Method

- We define $l(\boldsymbol{\beta}) \triangleq -\ln L(\mathbf{y}; \mathbf{X}, \boldsymbol{\beta})$ and find the maximum-likelihood estimate via

$$\hat{\boldsymbol{\beta}}_{\mathrm{ML}} = \arg\min_{\boldsymbol{\beta}} l(\boldsymbol{\beta})$$

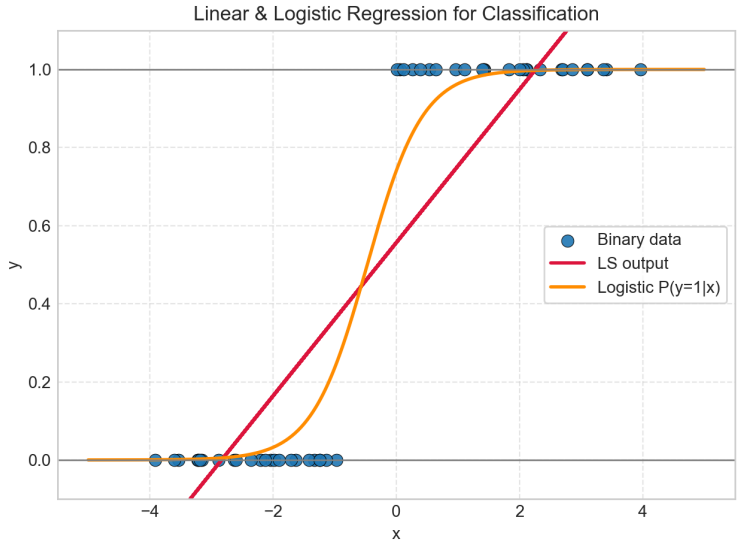
- Newton's method requires both the gradient and Hessian of the cost function $l(\boldsymbol{\beta})$.
- **Prediction:** Given a novel input $\mathbf{x}_*$, we can perform prediction in light of

$$P(y_* = 1; \mathbf{x}_*, \hat{\boldsymbol{\beta}}_{\mathrm{ML}}) = \frac{1}{1 + \exp(-\mathbf{x}_*^T \hat{\boldsymbol{\beta}}_{\mathrm{ML}})}$$

.

- Based on the above probability, we can further make a binary decision, i.e., choosing either 1 or 0, if necessary.

Comparison Between Linear and Logistic Regression



The End